

(Optional) Hands-on Lab: Advanced charts in Looker Studio

In this lab session, you'll be instructed on utilizing third-party tools for advanced visualization techniques. Following that, you'll delve into creating a bubble chart and word cloud specifically within Looker Studio.

Software used in this lab

Like the videos in the course, for the hands-on labs, we will be using Google's **Looker Studio** as this is available at no charge.

Data set used in this lab

The data set used in this lab comes from IBM Cognos Analytics. This data set is published by IBM. You can download the data set file directly from here: [CustomerLoyaltyProgram.csv](#) .

Objectives

After completing this lab, you will be able to:

- Utilize community visualizations.
- Create a word cloud.
- Create a scattered bubble chart.

Note: Sign in to Google Looker Studio and Make sure you have uploaded the dataset as shown in the previous labs. Properly uploading the dataset is essential for the successful completion of this lab exercise.

Exercise 1: Create a word cloud using community visualization tools

In this exercise, you will discover how to utilize community visualization tools to create various graphs that are not directly available in Looker Studio.

1. Start with the blank page and click on the **Community Visualization and Components**.

The screenshot displays the Google Looker Studio interface for an 'Untitled Report'. The top navigation bar includes 'File', 'Editing', 'View', 'Insert', 'Page', 'Arrange', 'Resource', and 'Help'. On the right, there are buttons for 'Reset', 'Share', 'View', and a menu icon. Below the navigation bar is a toolbar with icons for undo, redo, select, zoom, add page, add data, add chart, and add control. The 'Add chart' icon is highlighted with a red box. Below the toolbar is a search bar with the text '+ Add quick filter'. The main canvas area is empty. On the right side, there is a 'Data' panel showing a list of fields from the 'CustomerLoyaltyProgram (19).csv' dataset. The fields include City, Count, Country, Coupon Response, Customer Lifetime Value, Customer Name, Education, First Name, Gender, Income, Last Name, Latitude, Location Code, Longitude, Loyalty Count, Loyalty#, LoyaltyStatus, and Add a field. At the bottom of the 'Data' panel, there are buttons for 'Add a parameter' and 'Add Data'. A 'Community visualisations and components' panel is also visible on the right side of the canvas, showing a 'Let's get started' message and instructions on how to drag a field from the Data Panel to the canvas to add a new chart or select a component on the report canvas to edit it.

2. After selecting **Community Visualization and Components**, a popup will appear. Next, click **Explore More**.

The screenshot shows the Looker Studio interface with the 'Community visualisations' panel open on the left. The panel has a 'BETA' label and a description: 'Community visualisations are components built by third-party developers. Be sure to only add visualisations from trusted providers. [Learn more](#)'. Under the 'Featured' section, there are four visualisation options: Gauge (By Google), Radar Chart (By ClickInsight), Metric Funnel (By PowerMyAnalytics), and Sunburst Chart (By Supermetrics). At the bottom of this section, a red rectangular box highlights a button labeled '+ Explore more'. The main canvas on the right shows a 'Let's get started' message and a 'Data' panel on the far right listing fields from a 'CustomerLoyaltyProgram (19).csv' dataset, including City, Count, Country, Coupon Response, Customer Lifetime Value, Customer Name, Education, First Name, Gender, Income, Last Name, Latitude, Location Code, Longitude, Loyalty Count, Loyalty#, LoyaltyStatus, Marital Status, MonthsAsMember, Order Year, and Postal code. At the bottom of the Data panel are buttons for 'Add a field', 'Add a parameter', and 'Add Data'.

3. You will be redirected to the Community Gallery. Now, scroll down and search for the **Vega/Vega-Lite** option. Select it, and grant access by clicking the Allow button.

The screenshot shows the Looker Studio interface for an 'Untitled Report'. The top navigation bar includes 'File', 'Editing', 'View', 'Insert', 'Page', 'Arrange', 'Resource', and 'Help'. Below this is a toolbar with icons for undo, redo, pan, zoom, add page, add data, add chart, add control, and theme/layout. The main canvas is currently blank. On the right, the 'Data' panel shows a search bar and a list of fields from the 'CustomerLoyaltyProgram (19).csv' dataset: 'City', 'Count', 'Country', 'Coupon Response', and 'Customer Lifetime Value'. Below the main canvas is the 'Community Gallery' (marked as BETA), which displays a grid of chart templates. The 'Vega/Vega-Lite' chart, created by Jerry Chen, is highlighted with a red border. Other visible charts include Gauge, Sunburst, Heatmap, Sankey, Chord Diagram, Metric Funnel, Candlestick, Waterfall, CSV Filter Control, Range Slider, and Histogram.

4. After clicking on the **Vega/Vega-Lite**, a blank graph will appear on the canvas with your cursor. Click on the canvas to finalize and display the blank graph.

Untitled Report

File Editing View Insert Page Arrange Resource Help

Reset Share View ?

Add page Add data Add a chart Add a control Theme and layout

+ Add quick filter

Reset

Let's get started

Drag a field from the Data Panel to the canvas to add a new chart or select a component on the report canvas to edit it.

Click to add or drag and draw

Data

Search

CustomerLoyaltyProgram (19).csv

- City
- Count
- Country
- Coupon Response
- Customer Lifetime Value
- Customer Name
- Education
- First Name
- Gender
- Income
- Last Name
- Latitude
- Location Code
- Longitude
- Loyalty Count
- Loyalty#
- LoyaltyStatus
- Marital Status
- MonthsAsMember
- Order Year
- Postal code
- Add a field
- Add a parameter
- Add Data

5. By default, it will appear as shown in the screenshot below. We will proceed to adjust the dimensions and metrics.

Untitled Report

File Editing View Insert Page Arrange Resource Help

Reset Share View

Add page Add data Add a chart Add a control Theme and layout

+ Add quick filter

Reset

Community visualisation Properties

SET-UP STYLE

Data source

CustomerLoyaltyProgram (1...

+ BLEND DATA

Date Range Dimension

+ Add dimension

Fields

dimensions

RBC First Name

+ Add dimension

metrics

AUT Record Count

+ Add metric

Sort

AUT Record Count

Descending

Ascending

Filter

Community Visualisation Filter

Data

Search

CustomerLoyaltyProgram (19).csv

RBC City

123 Count

RBC Country

RBC Coupon Response

123 Customer Lifetime Value

RBC Customer Name

RBC Education

RBC First Name

RBC Gender

123 Income

RBC Last Name

123 Latitude

RBC Location Code

123 Longitude

123 Loyalty Count

123 Loyalty#

RBC LoyaltyStatus

RBC Marital Status

123 MonthsAsMember

123 Order Year

RBC Postal code

+ Add a field

+ Add a parameter

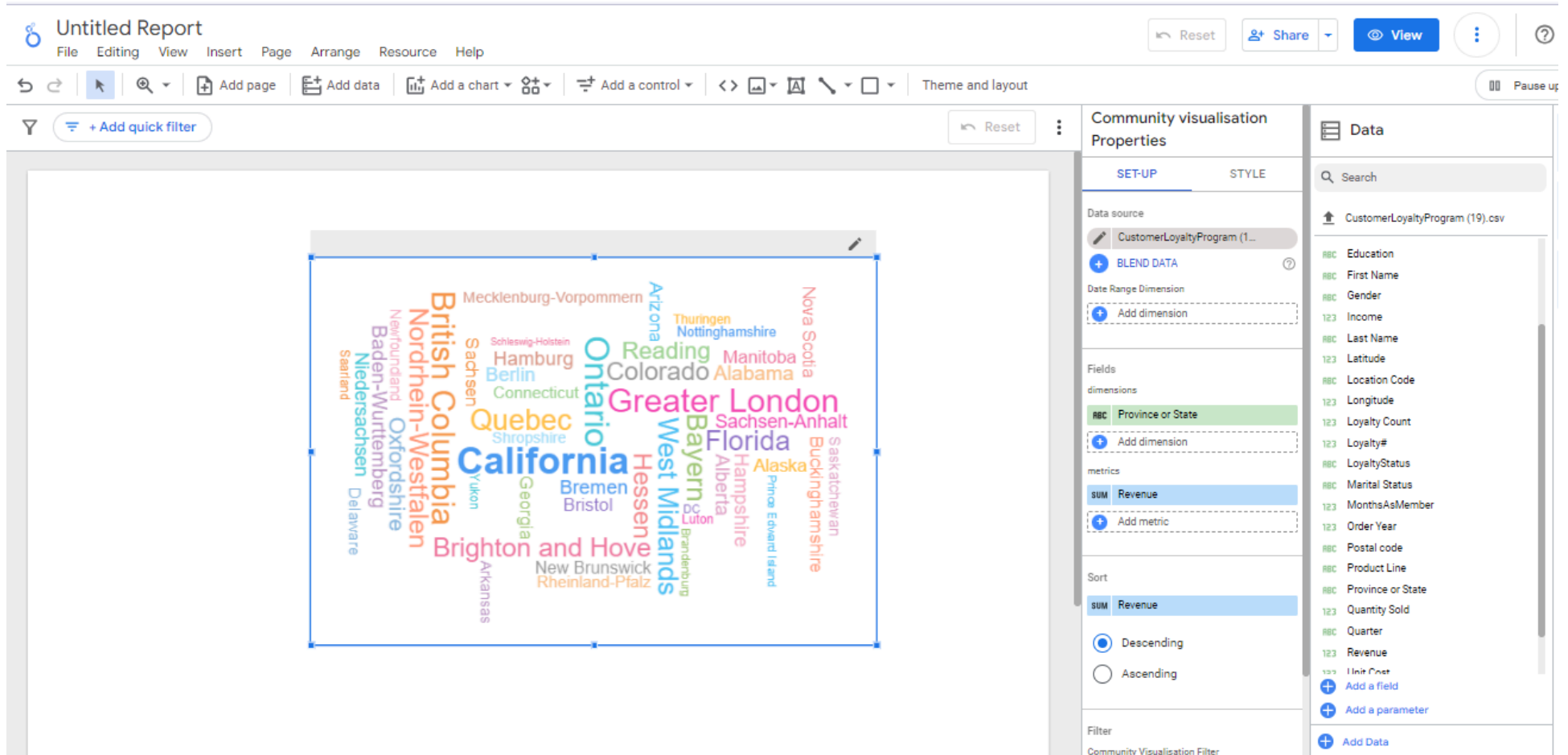
+ Add Data

6. Change the dimension from First Name to Province or State and metrics from Record count to Revenue.

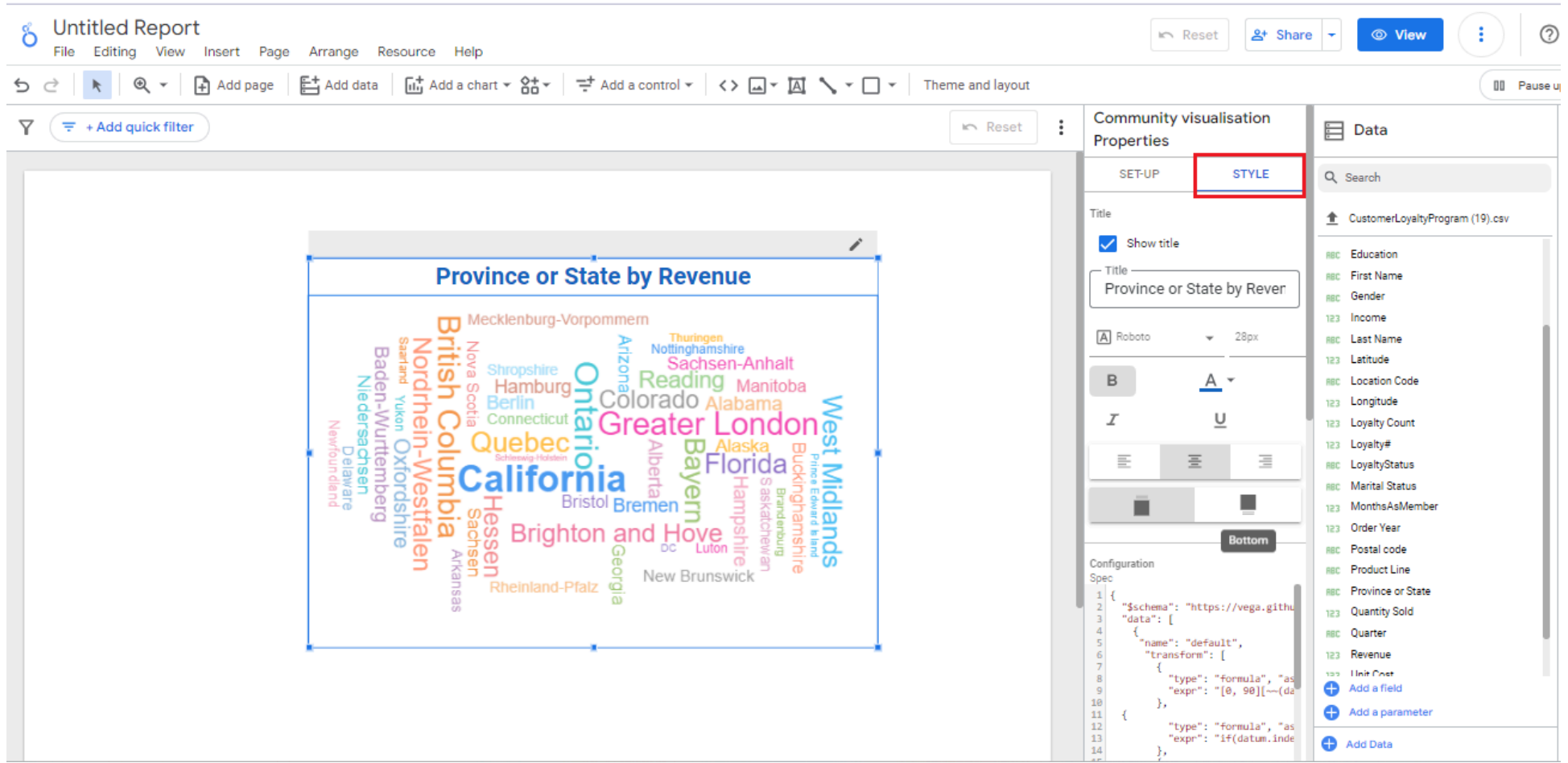
Dimension:

Metrics:





8. In the Style tab, set the title of the graph as **Province or State by Revenue** and align it in the center.



The above word cloud of **Province or State by Revenue** visualizes the relationship between different provinces or states and their corresponding revenue levels. In this visualization, the size of each province or state name within the cloud is proportional to its revenue. Typically, larger font sizes represent higher revenue values, while smaller font sizes represent lower revenue values. This type of visualization offers a quick and intuitive way to identify which provinces or states contribute the most to overall revenue and which ones contribute less.

Exercise 2: Create a Scatter bubble chart using multiple dimension

A scatter bubble chart with multiple dimensions is a powerful tool for visualizing data that has more than two dimensions. In a traditional scatter plot, you have two axes (X and Y) representing two variables, and you plot individual data points accordingly. However, in a scatter bubble chart with multiple dimensions, you can represent additional variables by using the size and color of the bubbles in addition to the X and Y coordinates.

Here, we'll create a visualization showcasing the relationship between **Product Line**, **LoyaltyStatus**, and **Quantity sold**.

1. To start, go to the toolbar and click **Add page**.
2. To add a new chart, click **Add a chart** and select a **scatter bubble chart**.

The screenshot shows the Looker Studio interface for an 'Untitled Report'. The top toolbar includes a search bar, navigation icons, and a 'Page 2 of 2' indicator. The 'Add a chart' button is highlighted with a red box. A dropdown menu is open, showing various chart types: Bar, Pie, Google Maps, Geo chart, Line, Area, Scatter, Pivot, and Bullet. The 'Scatter' category is expanded, and the 'Bubble chart' option is highlighted with a red box. The right sidebar shows the 'Data' panel with a search bar and a list of fields from the 'CustomerLoyaltyProgram (19).csv' dataset, including City, Count, Country, Coupon Response, Customer Lifetime Value, Customer Name, Education, First Name, Gender, Income, Last Name, Latitude, Location Code, Longitude, Loyalty Count, Loyalty#, LoyaltyStatus, and buttons to 'Add a field', 'Add a parameter', and 'Add Data'.

- Click on the canvas where you want it to be positioned. Click on the scatter bubble chart in the canvas, and then click **Properties**.
- Under the setup tab, we will change the dimensions and metrics to get the desired scatter bubble chart. From the data pane, drag **Product Line** to the Dimension field to replace **First Name**.

The screenshot shows the Looker Studio interface with the 'Chart' panel open. The 'SET-UP' tab is selected, and the 'Dimension' field is highlighted with a red arrow pointing to the 'Product Line' field in the 'Data' pane. The 'Metric X' field is 'Record Count' and the 'Metric Y' field is 'Loyalty#'. The 'Bubble Size Metric' is 'Latitude'. The 'Data' pane shows a list of fields from the 'CustomerLoyaltyProgram (19).csv' dataset, with 'Product Line' selected.

Field	Type
Last Name	RBC
Latitude	123
Location Code	RBC
Longitude	123
Loyalty Count	123
Loyalty#	123
LoyaltyStatus	RBC
Marital Status	RBC
MonthsAsMember	123
Order Year	123
Postal code	RBC
Product Line	RBC
Province or State	RBC
Quantity Sold	123
Quarter	RBC
Revenue	123
Unit Cost	123

- Now add one more dimension for **LoyaltyStatus**.

The screenshot displays the Looker Studio interface for configuring a chart. The left sidebar contains the 'Chart' panel with 'SET-UP' and 'STYLE' tabs. Under 'SET-UP', the 'Dimension' section lists 'Product Line' and 'LoyaltyStatus' (the latter is highlighted with a red arrow). The 'Metric X' section shows 'Record Count'. The 'Metric Y' section shows 'Loyalty#'. The 'Bubble Size Metric' section shows 'Latitude'. The 'Metric sliders' and 'Sort' sections are also visible. The 'Data' panel on the right shows a list of fields from 'CustomerLoyaltyProgram (19).csv', with 'LoyaltyStatus' highlighted. A red arrow points from the 'LoyaltyStatus' field in the 'Data' panel to the 'LoyaltyStatus' dimension in the 'Chart' panel. On the far right, there are buttons for 'Data', 'Properties', and 'Filter bar'.

6. Drag and drop the **Quantity Sold** into the **Bubble Size metric** area, replacing 'Latitude'.

☐ Theme and layout Pause updates

 Chart

SET-UP

STYLE

+ Add dimension

Dimension

RBC

Product Line

RBC

LoyaltyStatus

+ Add dimension

Drill down

Metric X

AUT

Record Count

Metric Y

SUM

Loyalty#

Bubble Size Metric

SUM

Quantity Sold

Metric sliders

Sort

 Data

Search

CustomerLoyaltyProgram (19).csv

RBC

Last Name

123

Latitude

RBC

Location Code

123

Longitude

123

Loyalty Count

123

Loyalty#

RBC

LoyaltyStatus

RBC

Marital Status

123

MonthsAsMember

123

Order Year

RBC

Postal code

RBC

Product Line

RBC

Province or State

123

Quantity Sold

RBC

Quarter

123

Revenue

123

Unit Cost

+ Add a field

+ Add a parameter

+ Add Data

 Data

 Properties

 Filter bar

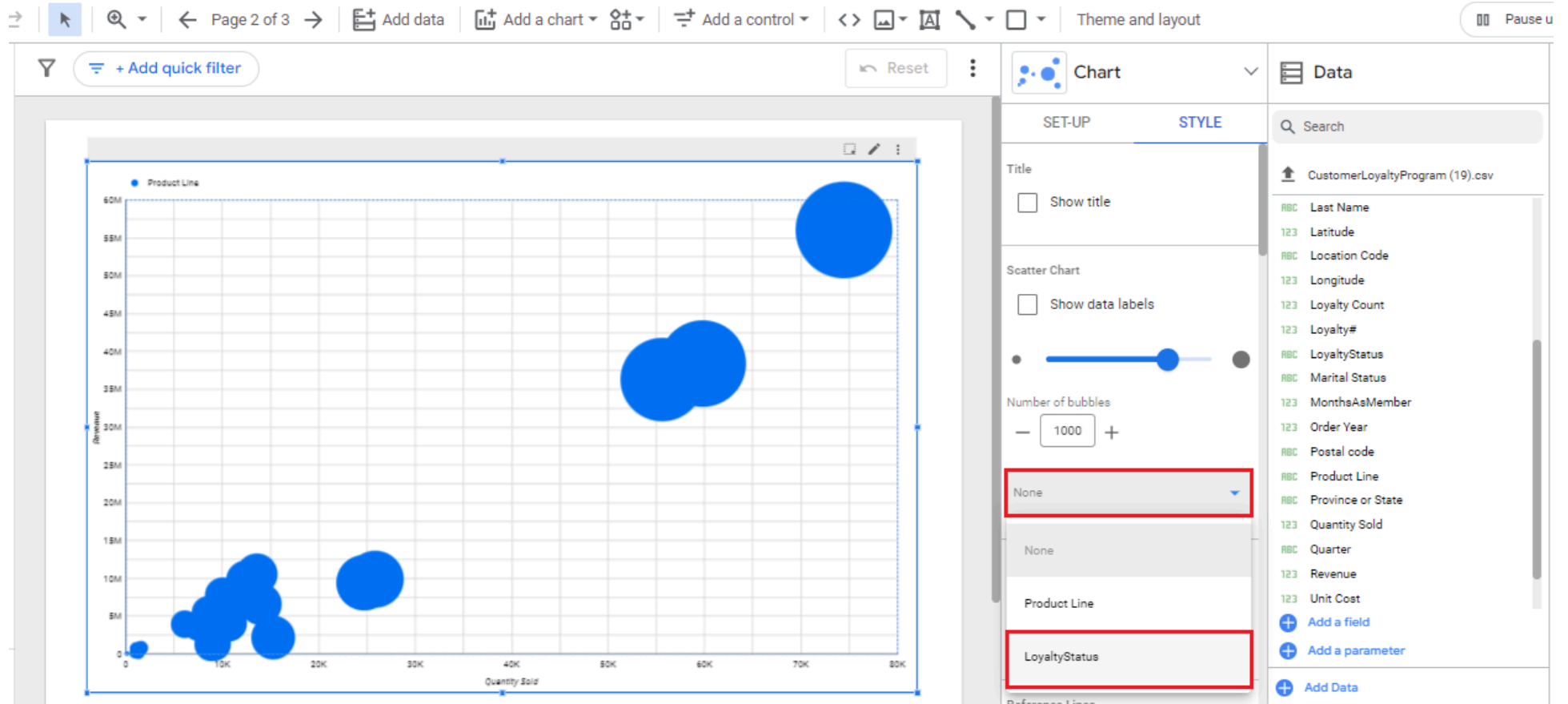
7. Now change the Metric X and Metric Y from **Record Count** and **Loyalty#** to **Quantity Sold** and **Revenue**.

<https://www.coursera.org/learn/data-visualization-dashboards-excel-cognos/supplement/40Xsv/optional-hands-on-lab-advanced-charts-in-looker-studio>

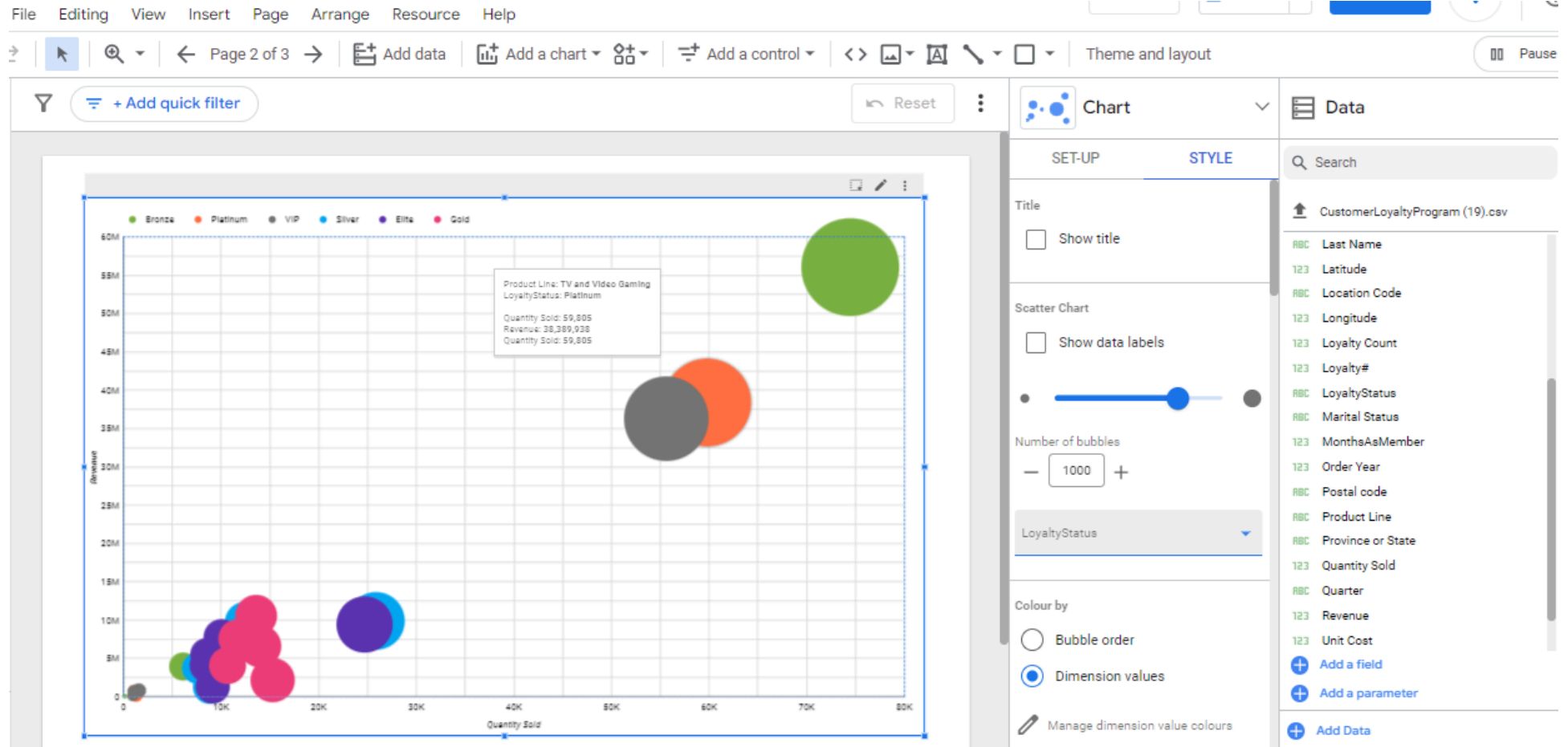
14/21

The screenshot displays the Looker Studio interface for configuring a chart. The left pane is titled 'Chart' and has two tabs: 'SET-UP' and 'STYLE'. The 'SET-UP' tab is active. Under the 'Dimension' section, 'Product Line' and 'LoyaltyStatus' are added. The 'Metric X' section has 'SUM Quantity Sold' selected, and the 'Metric Y' section has 'SUM Revenue' selected. Both metric selections are highlighted with red boxes. The 'Bubble Size Metric' section has 'SUM Quantity Sold' selected. The 'Drill down' and 'Metric sliders' options are toggled off. The right pane is titled 'Data' and shows a search bar and a list of fields from the 'CustomerLoyaltyProgram (19).csv' dataset. The fields listed are: Last Name, Latitude, Location Code, Longitude, Loyalty Count, Loyalty#, LoyaltyStatus, Marital Status, MonthsAsMember, Order Year, Postal code, Product Line, Province or State, Quantity Sold, Quarter, Revenue, and Unit Cost. Red arrows point from the 'SUM Quantity Sold' and 'SUM Revenue' metrics in the 'Chart' pane to the 'Quantity Sold' and 'Revenue' fields in the 'Data' pane, respectively.

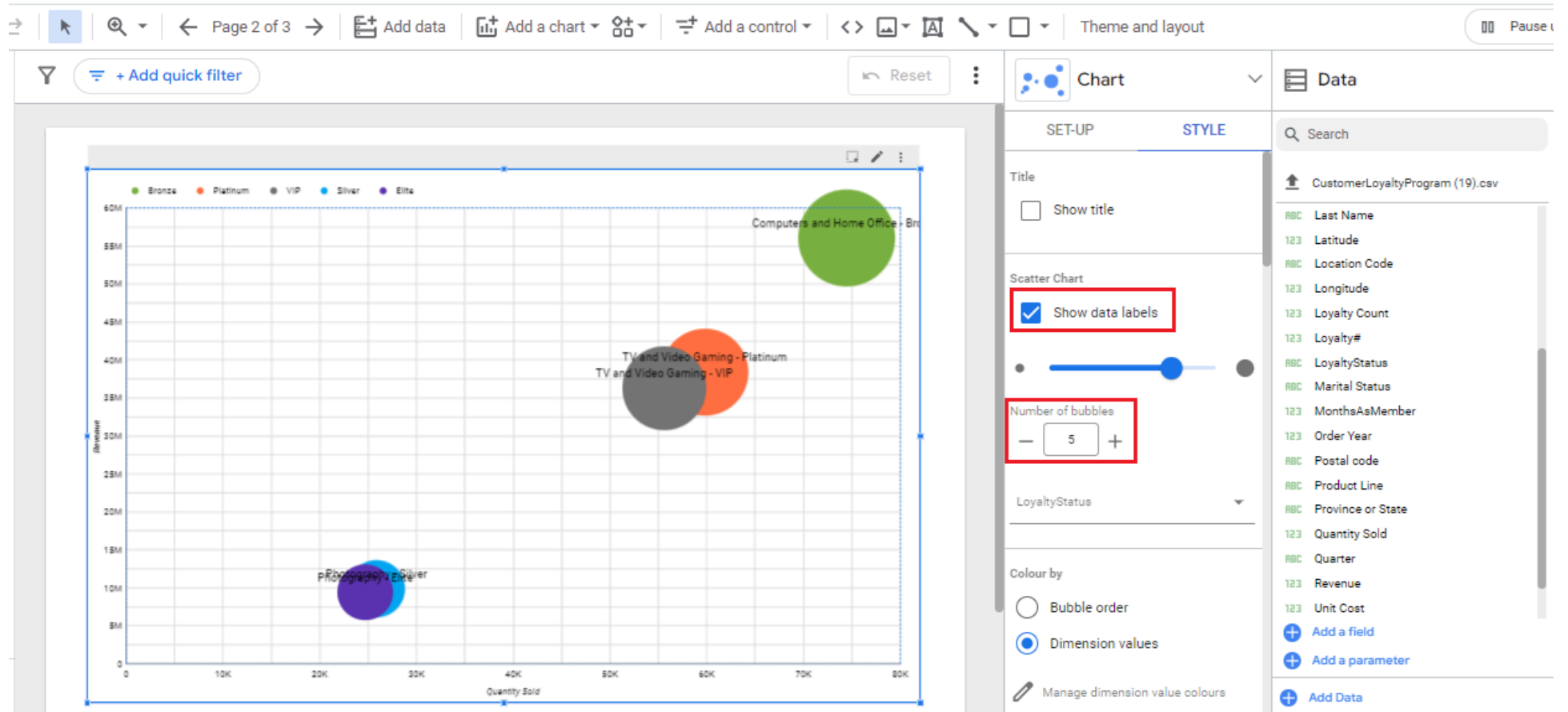
8. We will get the initial graph as shown below. Next, click the **STYLE** tab in the chart's Properties pane, then click on the drop down of **Bubble Color** field and select **Loyalty status**.



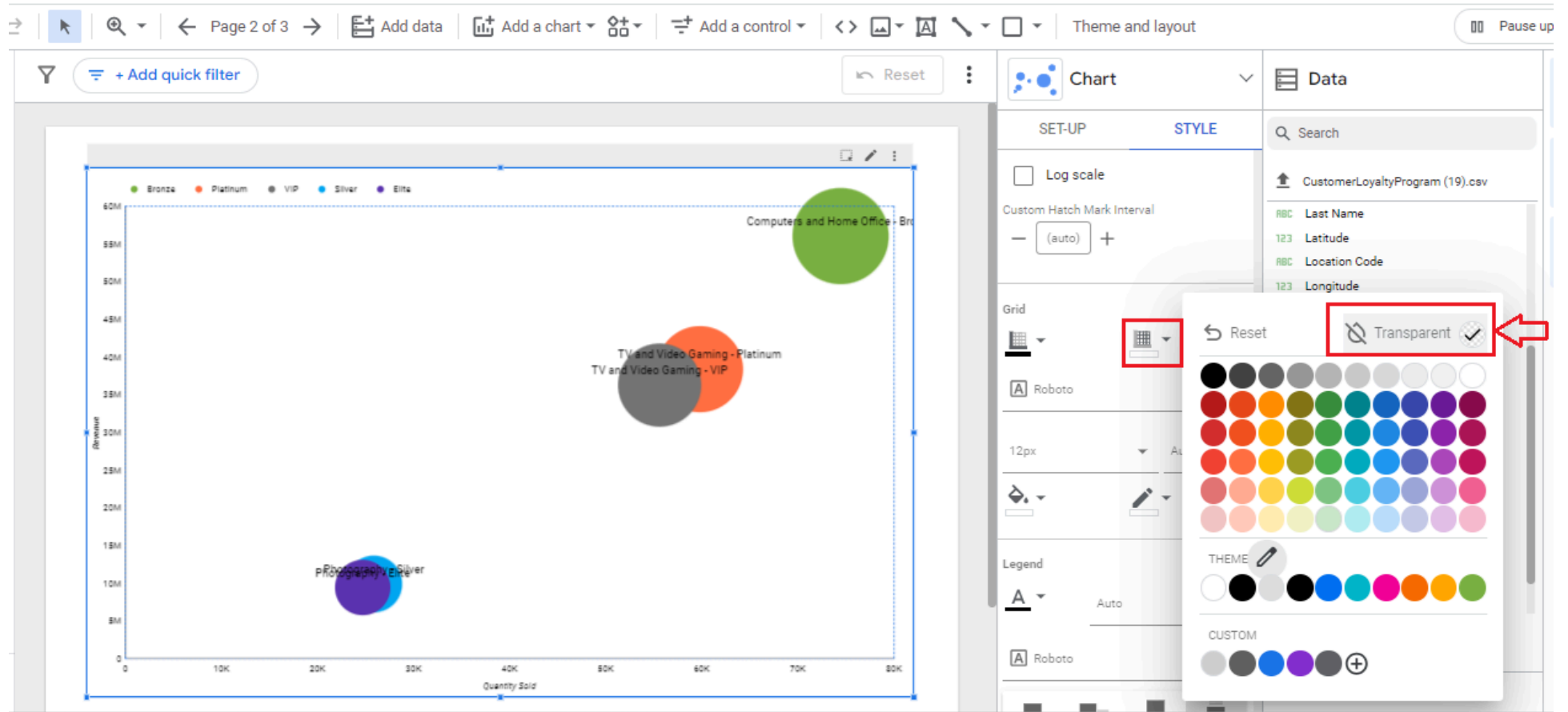
9. The graph will appear like the one below after setting the color. If you hover over the bubbles, you can see each bubble's dimensions and metrics in a dialog box, such as Loyalty status, Quantity sold, Revenue, and Product line.



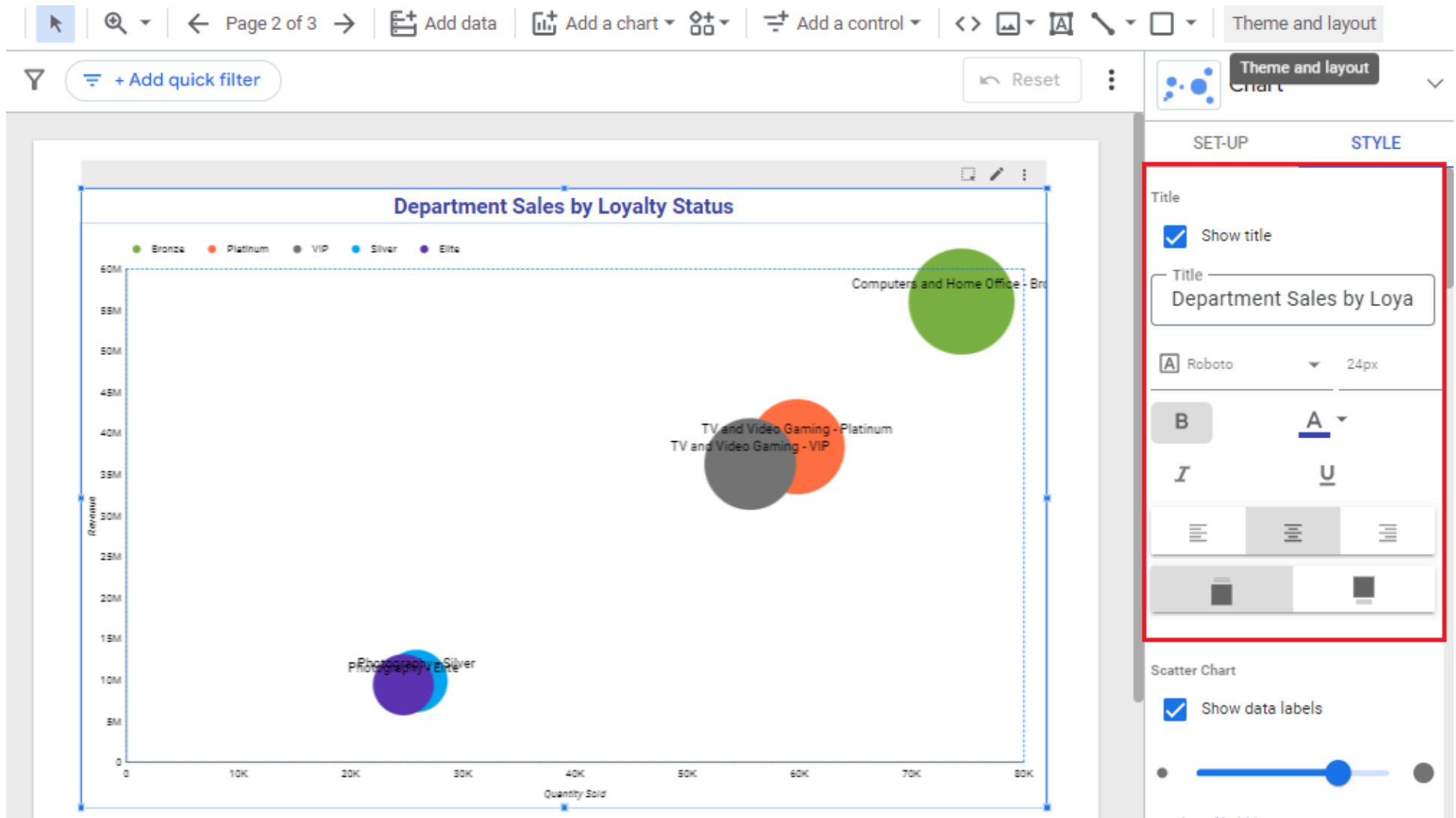
10. From the Style tab, set the **Number of bubbles** from 1000 to 5 and check the **Show data labels**.



11. Next, in the STYLE tab in the chart's Properties pane, go to the **Grid** settings, select the **Grid colour**, and check the **Transparent** box.



12. Check the Show Title box and type the title as **Department Sales by Loyalty Status**. Style the text as **24pt, bold, and dark blue**. Align it with the center of the chart visualization.



The scatter bubble graph, using dimensions of Product Line and Loyalty Status, and metrics of Quantity Sold (X-axis), Revenue (Y-axis), with bubble size representing Quantity Sold and bubble color indicating Loyalty Status, provides a comprehensive visualization of sales data. Each bubble corresponds to a product line, with its position on the X-axis showing the quantity sold and its position on the Y-axis showing the total revenue generated. The size of the bubble reflects the quantity sold, with larger bubbles indicating higher sales volumes. The color of the bubble represents the loyalty status of the customers, allowing for easy differentiation between various customer segments, such as Gold, Silver, or Bronze. This graph helps in understanding the performance of different product lines, the impact of customer loyalty on sales and revenue, and the relationship between the quantity sold and the revenue generated.

Congratulations! You have completed this hands-on lab and you are now ready for the next topic.

For more help, you can refer to the [Tutorial on Looker Studio by Google](#) .